Substantive Citation Report

Twenty-one substantive citing articles, each published in an ICORE2026 A*-rated conference, cited more than 100 times on Google Scholar, or using or extending TQT in its own methods or experiments. Articles are ordered by the documented use of TQT: first those that adopt, implement or extend it in their own methods or experiments, then those that compare with it, credit it or state its use in a single sentence, then those that discuss it or apply it through a modified implementation or a vendor toolchain. Within each group, the depth of reliance, venue standing and the independence of the citing authors determine the order.

21
selected articles

108
marked discussion pages

30
countries / territories in the wider survey

Selection and source documentation

The selection basis lists every criterion an article meets. ICORE2026 A* venue: published in a conference rated A* in ICORE2026; ACM Transactions on Graphics 39(6) contains the SIGGRAPH Asia 2020 technical papers, and CVPR workshops carry no separate rating. >100 Scholar citations: more than 100 Google Scholar citations in the saved snapshot. Uses or extends TQT: the article's own methods or experiments use, follow or extend TQT, or compare an implemented extension against it, as stated in the source passages; background mentions, comparisons without such use and statements that a design is merely similar to TQT do not qualify. Counts belong to the citing articles' Scholar records or version clusters; two records show no cited-by count in the saved snapshot. Each entry identifies its reviewed publication version, the citing authors' affiliations, any employer link and the contribution supported by the source passage.

Authorship links and documented scholarly use

Each article identifies the cited work and authors, the specific research connection and the original discussion pages. The bibliography link establishes authorship of the corresponding work by Sambhav Jain and coauthors.

Orange rectangles mark selected discussion, citations and TQT results; blue rectangles identify the corresponding bibliography entries. The marked packet follows the article order in this report and retains original page numbers. Article numbers identify the corresponding marked files.

> **International citation survey.** The wider survey maps 219 citing works to 232 institution/location entries across 30 countries/territories. These totals describe the broader citation network, separately from this report’s selection.

Report PDF · Marked discussions · Interactive map

Snapshot: October 7, 2026. Sambhav Jain’s Google Scholar profile. Counts belong to the saved citing-article records or version clusters.

Selected citing articles (21)

No.	Article / venue	Affiliations	Scholar	Selection basis
01	**Understanding How Orthogonality of Parameters Improves Quantization of Neural Networks** IEEE Transactions on Neural Networks and Learning Systems 34(12), 10737–10746 (2023) · Marked PDF	NVIDIA Corporation Bilkent University	12	Uses or extends TQT
02	**Mixed Precision DNNs: All you need is a good parametrization** ICLR 2020 · Marked PDF	Sony Europe B.V. Sony Corporate	230	ICORE2026 A* venue >100 Scholar citations Uses or extends TQT
03	**A Reduced-Precision Network for Image Reconstruction** ACM Transactions on Graphics 39(6), Article 231 (SIGGRAPH Asia 2020) · Marked PDF	University of California, Santa Cruz Intel Corporation	41	ICORE2026 A* venue Uses or extends TQT
04	**Optimizing the Deployment of Tiny Transformers on Low-Power MCUs** IEEE Transactions on Computers 74(2), 526–541 (2025) · Marked PDF	ETH Zürich University of Bologna Politecnico di Torino	47	Uses or extends TQT
05	**Understanding and Overcoming the Challenges of Efficient Transformer Quantization** EMNLP 2021 · Marked PDF	Qualcomm AI Research	271	ICORE2026 A* venue >100 Scholar citations Uses or extends TQT
06	**7 µJ/inference end-to-end gesture recognition from dynamic vision sensor data using ternarized hybrid convolutional neural networks** Future Generation Computer Systems 149 (2023), 717–731 · Marked PDF	ETH Zürich University of Bologna	12	Uses or extends TQT
07	**Integer-Only CNNs with 4 Bit Weights and Bit-Shift Quantization Scales at Full-Precision Accuracy** Electronics 10(22), 2823 (2021) · Marked PDF	KU Leuven	15	Uses or extends TQT
08	**A Resource Efficient Integer-Arithmetic-Only FPGA-Based CNN Accelerator for Real-Time Facial Emotion Recognition** IEEE Access 9 (2021), 104367–104381 · Marked PDF	Inha University Sangmyung University	54	Uses or extends TQT
09	**A Low Memory Footprint Quantized Neural Network for Depth Completion of Very Sparse Time-of-Flight Depth Maps** CVPR Workshops 2022 (CVPRW), 2686–2695 · Marked PDF	EPFL Sony Depthsensing Solutions NV Sony Europe B.V. University of Padova	12	Uses or extends TQT

No.	Article / venue	Affiliations	Scholar	Selection basis
10	**QKD: Quantization-aware Knowledge Distillation** arXiv preprint, 2019 · Marked PDF	Seoul National University Qualcomm AI Research	125	>100 Scholar citations Uses or extends TQT
11	**Improving Low-Precision Network Quantization via Bin Regularization** ICCV 2021, 5241–5250 · Marked PDF	Xilinx Inc.	65	ICORE2026 A* venue Uses or extends TQT
12	**Bayesian Bits: Unifying Quantization and Pruning** NeurIPS 2020 · Marked PDF	Qualcomm AI Research	207	ICORE2026 A* venue >100 Scholar citations
13	**F8Net: Fixed-Point 8-bit Only Multiplication for Network Quantization** ICLR 2022 · Marked PDF	Snap Inc. Northeastern University Rice University	72	ICORE2026 A* venue
14	**Accelerating configuration of Reconfigurable Intelligent Surfaces through a hardware-enhanced deep learning approach** Computers and Electrical Engineering 134 (2026), 111101 · Marked PDF	University of Granada Safran Electronics & Defense University of Málaga Biomedic Research Institute of Málaga (IBIMA)	None shown	Uses or extends TQT
15	**FELIX: A Ferroelectric FET Based Low Power Mixed-Signal In-Memory Architecture for DNN Acceleration** ACM Transactions on Embedded Computing Systems 21(6), Article 84 (2022) · Marked PDF	Robert Bosch GmbH Fraunhofer IPMS University of Kaiserslautern	50	Uses or extends TQT
16	**HMQ: Hardware Friendly Mixed Precision Quantization Block for CNNs** ECCV 2020 · Marked PDF	Sony Semiconductor Israel	95	ICORE2026 A* venue
17	**Up or Down? Adaptive Rounding for Post-Training Quantization** ICML 2020 · Marked PDF	Qualcomm AI Research	1,224	ICORE2026 A* venue >100 Scholar citations
18	**A White Paper on Neural Network Quantization** arXiv / Qualcomm white paper, 2021 · Marked PDF	Qualcomm AI Research	1,362	>100 Scholar citations
19	**Bringing AI to edge: From deep learning's perspective** Neurocomputing 485 (2022), 297–320 · Marked PDF	Nanyang Technological University HP Inc.	247	>100 Scholar citations
20	**INT8 Activation Ternary or Binary Weights Networks: Unifying Between INT8 and Lower-bit Width Quantization** Journal of Robotics, Networking and Artificial Life 9(2) (2022), 171–176 · Marked PDF	Kyushu Institute of Technology	None shown	Uses or extends TQT
21	**Efficient SAR Vessel Detection for FPGA-Based On-Satellite Sensing** SEC '25, Tenth ACM/IEEE Symposium on Edge Computing (2025) · Marked PDF	The Alan Turing Institute	6	Uses or extends TQT

SELECTED CITING ARTICLES

ARTICLE 01 · SUBSTANTIVE CITATION REPORT

Understanding How Orthogonality of Parameters Improves Quantization of Neural Networks

Sukru Burc Eryilmaz; Aysegul Dundar
IEEE Transactions on Neural Networks and Learning Systems 34(12), 10737–10746 (2023) · 12 Google Scholar citations

Uses or extends TQT

Documented contribution: Direct methodological adoption

NVIDIA and Bilkent University researchers adopt TQT as their quantized retraining method, stating "we use TQT [25] as our baseline retraining method" (p. 5) and crediting it with "state-of-the-art results for several networks" (p. 1). They train thresholds in the log domain and follow the five-epoch schedule of [25], noting implementation differences (pp. 4–6), report accuracy before and after TQT (Table I, p. 6), contrast results without TQT (p. 7) and in the conclusion describe "a state-of-the-art TQT quantized retraining method" (p. 9).

Authorship link: Jain, Gural, Wu and Dick (2019), Trained quantization thresholds for accurate and efficient fixed-point inference of deep neural networks. Bibliography p. 10.

Reviewed version: Final IEEE TNNLS print, vol. 34, no. 12. **Pinpoint:** Physical PDF pp. 1–9 (printed pp. 10737–10745): TQT introduced as the quantized retraining method, log-domain threshold training as described in [25], TQT baseline retraining, experiments with and without TQT (Fig. 1(c), Table I, Section V-C, Fig. 3) and the conclusion; bibliography p. 10 (printed p. 10746), reference [25].
Count: E-0402, `data.organic_results[19].inline_links.cited_by.total`. Venue: E-1429.

Marked file: 01-Orthogonality-MARKED.pdf

Original article · Scholar cited-by list

ARTICLE 02 · SUBSTANTIVE CITATION REPORT

Mixed Precision DNNs: All you need is a good parametrization

Stefan Uhlich; Lukas Mauch; Fabien Cardinaux; Kazuki Yoshiyama; Javier Alonso García; Stephen Tiedemann; Thomas Kemp; Akira Nakamura
ICLR 2020 · 230 Google Scholar citations

ICORE2026 A* venue · >100 Scholar citations · Uses or extends TQT

Documented contribution: Implemented experimental benchmark

Sony researchers implement TQT as the fixed-bitwidth benchmark for their mixed-precision method, learning qmax at a fixed 2-bit width "as was done in (Jain et al., 2019)" and reporting the "TQT" configuration in their CIFAR-10 and ImageNet tables (pp. 8–9). They credit Jain et al., in work parallel to their own, with learning the quantizer's dynamic range through the straight-through estimator (p. 2) and cite it for the power-of-two step size that efficient hardware requires (p. 4).

Authorship link: Jain, Gural, Wu and Dick (2019), Trained uniform quantization for accurate and efficient neural network inference on fixed-point hardware. Bibliography p. 10.

Reviewed version: Author-hosted arXiv copy bearing ICLR 2020 publication header. **Pinpoint:** PDF/printed pp. 1–2, 4–5, 8–9, introduction, quantizer constraints, experimental setup and TQT result rows; bibliography p. 10.
Count: E-0430, `data.organic_results[11].inline_links.cited_by.total`. Venue: E-0328; E-1494; E-1495.

Marked file: 02-Mixed-Precision-DNNs-MARKED.pdf

Original article · Reviewed arXiv version · Scholar cited-by list

SELECTED CITING ARTICLES

ARTICLE 03 · SUBSTANTIVE CITATION REPORT

A Reduced-Precision Network for Image Reconstruction

Manu Mathew Thomas; Karthik Vaidyanathan; Gabor Liktor; Angus G. Forbes
ACM Transactions on Graphics 39(6), Article 231 (SIGGRAPH Asia 2020) · 41 Google Scholar citations

ICORE2026 A* venue | Uses or extends TQT

Documented contribution: Direct methodological adoption

University of California, Santa Cruz and Intel researchers adopt TQT's threshold training in the quantization of their 4-bit QW-Net image-reconstruction network, stating "Following Jain et al. [2019], we train the thresholds in log space" to improve stability with quantized training (Section 5.1, p. 6). The article explains that Jain et al. and Esser et al. showed that quantization thresholds could be trained to improve accuracy, with a straight-through-estimator gradient that lets thresholds grow for dynamic range or shrink for precision (p. 3). The log-space training is attributed to Jain et al. alone.

Authorship link: Jain, Gural, Wu and Dick (2019), Trained quantization thresholds for accurate and efficient fixed-point inference of deep neural networks. Bibliography p. 11.

Reviewed version: ACM Transactions on Graphics publisher PDF. **Pinpoint:** Physical PDF pp. 3 and 6 (Section 5.1): trained quantization thresholds and log-space threshold training following Jain et al. [2019]; bibliography p. 11.
Count: E-0379, `data.organic_results[0].inline_links.cited_by.total`. Venue: E-1459; E-1460.

Marked file: 03-QW-Net-MARKED.pdf

Original article · Scholar cited-by list

ARTICLE 04 · SUBSTANTIVE CITATION REPORT

Optimizing the Deployment of Tiny Transformers on Low-Power MCUs

Victor J.B. Jung; Alessio Burrello; Moritz Scherer; Francesco Conti; Luca Benini
IEEE Transactions on Computers 74(2), 526–541 (2025) · 47 Google Scholar citations

Uses or extends TQT

Documented contribution: Direct methodological use

ETH Zürich, University of Bologna and Politecnico di Torino researchers quantize their tiny transformers for low-power microcontrollers with TQT, stating "We use the uniform symmetric quantizer from the SotA [state-of-the-art] Trained Quantization Threshold (TQT) method [3]", and specify the functional behavior of the TQT quantizer in their deployment flow (Section IV-C, p. 6). The introduction also cites TQT for data quantization among standard hardware-agnostic optimization techniques (p. 1).

Authorship link: Jain, Gural, Wu and Dick (2020), Trained quantization thresholds for accurate and efficient fixed-point inference of deep neural networks. Bibliography p. 15.

Reviewed version: Published IEEE Transactions on Computers print. **Pinpoint:** Physical PDF p. 1 (printed p. 526), Section I: data quantization citation [3]; p. 6 (printed p. 531), Section IV-C: the uniform symmetric TQT quantizer used in the QuantLib flow; bibliography p. 15 (printed p. 540), reference [3].
Count: E-0375, `data.organic_results[6].inline_links.cited_by.total`. Venue: E-1421.

Marked file: 04-Tiny-Transformers-MARKED.pdf

Original article · Scholar cited-by list

SELECTED CITING ARTICLES

ARTICLE 05 · SUBSTANTIVE CITATION REPORT

Understanding and Overcoming the Challenges of Efficient Transformer Quantization

Yelysei Bondarenko; Markus Nagel; Tijmen Blankevoort
EMNLP 2021 · 271 Google Scholar citations

ICORE2026 A* venue | >100 Scholar citations | Uses or extends TQT

Documented contribution: Methodological adaptation

Qualcomm researchers adapt the procedure of Jain et al. and Esser et al. to learn the ranges of both weights and activations in their quantization-aware training of BERT-like transformer models (p. 7), having noted that such ranges can be learned jointly with the weights during training "as in Esser et al. (2019); Jain et al. (2019)" (p. 3).

Authorship link: Jain, Gural, Wu and Dick (2019), Trained uniform quantization for accurate and efficient neural network inference on fixed-point hardware. Bibliography p. 11.

Reviewed version: Published proceedings PDF. **Pinpoint:** Physical PDF pp. 3, 7, learnable weight/activation ranges and their adaptation to transformers; bibliography p. 11 / printed p. 7957.
Count: E-0430, `data.organic_results[6].inline_links.cited_by.total`. Venue: E-0326.

Marked file: 05-Transformer-Quantization-MARKED.pdf

Original article · Scholar cited-by list

ARTICLE 06 · SUBSTANTIVE CITATION REPORT

7 µJ/inference end-to-end gesture recognition from dynamic vision sensor data using ternarized hybrid convolutional neural networks

Georg Rutishauser; Moritz Scherer; Tim Fischer; Luca Benini
Future Generation Computer Systems 149 (2023), 717–731 · 12 Google Scholar citations

Uses or extends TQT

Documented contribution: Direct methodological use and comparison

ETH Zürich and University of Bologna researchers train their ternarized gesture-recognition networks with two quantization-aware training algorithms, INQ and TQT (p. 6). They use TQT to learn weight and activation clipping bounds (pp. 5–6), specify its hyperparameters (Table 3, p. 9) and train their Pareto-front networks with it (p. 9). They report that "the combination of ReLU activations and TQT yields the highest classification accuracy", ahead of INQ (Table 4, p. 10).

Authorship link: Jain, Gural, Wu and Dick (2020), Trained quantization thresholds for accurate and efficient fixed-point inference of deep neural networks. Bibliography p. 14.

Reviewed version: Published open-access journal PDF (Elsevier). **Pinpoint:** Physical PDF pp. 5–6, 9–10: TQT as one of two quantization-aware training algorithms, TQT hyperparameters and Table 4 TQT versus INQ results; bibliography p. 14, reference [67].
Count: E-0382, `data.organic_results[9].inline_links.cited_by.total`.

Marked file: 06-DVS-Gesture-MARKED.pdf

Original article · Scholar cited-by list

SELECTED CITING ARTICLES

ARTICLE 07 · SUBSTANTIVE CITATION REPORT

Integer-Only CNNs with 4 Bit Weights and Bit-Shift Quantization Scales at Full-Precision Accuracy

Maarten Vandersteegen; Kristof Van Beeck; Toon Goedemé
Electronics 10(22), 2823 (2021) · 15 Google Scholar citations

Uses or extends TQT

Documented contribution: Direct methodological adoption and extension

KU Leuven researchers take TQT as the starting point of their work: crediting Jain et al. with having "already proved" that 8-bit models with power-of-two scales reach full-precision accuracy, they set out to extend this to 4-bit weights (pp. 2, 14). They adopt its method, optimizing power-of-two scales "as proposed by Jain et al." (p. 12) with its quantizer, hyperparameters and scale freezing (p. 17), and extend its freezing to per-channel scales, finding its "per-layer results were consistent with ours" (p. 19). Table 2 lists TQT as the only prior power-of-two-scale method, within 1.6 points of full precision at 4-bit weights (p. 20).

Authorship link: Jain, Gural, Wu and Dick (2020), Trained quantization thresholds for accurate and efficient fixed-point inference of deep neural networks. Bibliography p. 23.

Reviewed version: Published journal PDF from KU Leuven repository. **Pinpoint:** PDF/printed pp. 2, 11–14, 17–20: Jain et al. [12] in the introduction, quantizer design, adopted training practices and TQT Table 2 results; bibliography p. 23, reference [12].
Count: E-0459, `data.organic_results[0].inline_links.cited_by.total`.

Marked file: 07-Integer-Only-CNNs-MARKED.pdf

Original article · Scholar cited-by list

ARTICLE 08 · SUBSTANTIVE CITATION REPORT

A Resource Efficient Integer-Arithmetic-Only FPGA-Based CNN Accelerator for Real-Time Facial Emotion Recognition

Jaemyung Kim; Jin-Ku Kang; Yongwoo Kim
IEEE Access 9 (2021), 104367–104381 · 54 Google Scholar citations

Uses or extends TQT

Documented contribution: Method extension and implemented comparison

Inha University and Sangmyung University researchers identify TQT and LSQ as "the current state-of-the-art quantization methods" (p. 6) and develop log level threshold quantization (LLTQ) from TQT's power-of-two scale mapping: LLTQ maps the trainable parameter directly to a power-of-two scale and replaces TQT's pre-calibration with first-batch statistics (pp. 3–4, 6). They illustrate and train TQT, LSQ and LLTQ (pp. 7, 10) and report that TQT, unlike LSQ, is "hardware-friendly" (pp. 11–12).

Authorship link: Jain, Gural, Wu and Dick (2020), Trained quantization thresholds for accurate and efficient fixed-point inference of deep neural networks. Bibliography p. 15.

Reviewed version: IEEE Access open-access publisher PDF. **Pinpoint:** Physical PDF pp. 3–4, 6–7, 10–12: TQT in related work, LLTQ proposed to address the disadvantages of previous works including TQT and derived as an alternative to TQT's power-of-two scale mapping, Figure 3, the implemented TQT/LSQ/LLTQ comparison and its Table 1 TQT row; bibliography p. 15, reference [39].
Count: E-0393, `data.organic_results[0].inline_links.cited_by.total`.

Marked file: 08-LLTQ-MARKED.pdf

Original article · Scholar cited-by list

SELECTED CITING ARTICLES

ARTICLE 09 · SUBSTANTIVE CITATION REPORT

A Low Memory Footprint Quantized Neural Network for Depth Completion of Very Sparse Time-of-Flight Depth Maps

Xiaowen Jiang; Valerio Cambareri; Gianluca Agresti; Cynthia Ifeyinwa Ugwu; Adriano Simonetto; Fabien Cardinaux; Pietro Zanuttigh
CVPR Workshops 2022 (CVPRW), 2686–2695 · 12 Google Scholar citations

Uses or extends TQT

Documented contribution: Direct methodological use

EPFL, Sony and University of Padova researchers report that their most effective uniform-precision quantization-aware training procedure initializes from the pretrained float32 model and uses "learnable qmax as in the Trained Quantization Threshold (TQT) approach [23]", stabilized with cosine learning-rate decay and RMSprop followed by Adam (Section 5.1, p. 5). The related-work section cites TQT among recent quantization-aware training methods that "proved effective on classification tasks" (p. 2).

Authorship link: Jain, Gural, Wu and Dick (2020), Trained quantization thresholds for accurate and efficient fixed-point inference of deep neural networks. Bibliography p. 9.

Reviewed version: IEEE Xplore published print, CVPRW 2022 proceedings. **Pinpoint:** Physical PDF p. 2, Section 2.2 (printed p. 2687): QAT citation group including [23]; p. 5, Section 5.1 (printed p. 2690): uniform-precision QAT using learnable qmax as in TQT [23]; bibliography p. 9 (printed p. 2694), reference [23].
Count: E-0402, `data.organic_results[14].inline_links.cited_by.total`. Venue: E-1422.

Marked file: 09-Depth-Completion-MARKED.pdf

Original article · Scholar cited-by list

ARTICLE 10 · SUBSTANTIVE CITATION REPORT

QKD: Quantization-aware Knowledge Distillation

Jangho Kim; Yash Bhalgat; Jinwon Lee; Chirag Patel; Nojun Kwak
arXiv preprint, 2019 · 125 Google Scholar citations

>100 Scholar citations Uses or extends TQT

Documented contribution: Use in the baseline implementation

Seoul National University and Qualcomm researchers state that TQT [17] and LSQ [8] "introduce uniform quantization using trainable interval values" and that "Our QKD leverages these trainable approaches in the baseline quantization implementation" (p. 2), introducing trainable interval parameters for each layer's weights and input activations similar to [8, 17, 45] (p. 3). They also cite TQT on the quantization sensitivity of depthwise convolutions (p. 7).

Authorship link: Jain, Gural, Wu and Dick (2019), Trained quantization thresholds for accurate and efficient fixed-point inference of deep neural networks. Bibliography p. 9.

Reviewed version: Reviewed arXiv manuscript. **Pinpoint:** Physical PDF pp. 2–3, 7, trainable intervals, dequantization and depthwise-layer sensitivity; bibliography p. 9, reference [17].
Count: E-0378, `data.organic_results[0].inline_links.cited_by.total`.

Marked file: 10-QKD-MARKED.pdf

Original article · Scholar cited-by list

SELECTED CITING ARTICLES

ARTICLE 11 · SUBSTANTIVE CITATION REPORT

Improving Low-Precision Network Quantization via Bin Regularization

Tiantian Han; Dong Li; Ji Liu; Lu Tian; Yi Shan
ICCV 2021, 5241–5250 · 65 Google Scholar citations

ICORE2026 A* venue | Uses or extends TQT

Documented contribution: Extension of the TQT baseline

Xilinx researchers in Beijing build bin regularization on a hardware-friendly quantization baseline that they re-implement following references [19, 21], where [19] is TQT: power-of-two step sizes, low-bit first and last layers, quantized inputs and outputs of element-wise additions, 8-bit biases and batch normalization merged during training (p. 8). Table 9 reports 63.8% top-1 accuracy for TQT, 65.8% for TQT+L2 and 67.5% for TQT+BR on 4-bit MobileNetV2, which the text describes as "+3.7% over TQT" (p. 8). The authors are affiliated with Xilinx Inc.; the entry is employer-linked.

Authorship link: Jain, Gural, Wu and Dick (2020), Trained quantization thresholds for accurate and efficient fixed-point inference of deep neural networks. Bibliography p. 9.

Reviewed version: IEEE Xplore published print, ICCV 2021 proceedings. **Pinpoint:** Physical PDF p. 2 (printed p. 5242): related-work QAT citation group including [19]; p. 8 (printed p. 5248): Section 4.8 hardware-friendly baseline following [19, 21], Table 9 TQT, TQT+L2 and TQT+BR rows and the reported +3.7% over TQT; bibliography p. 9 (printed p. 5249), reference [19].
Count: E-0391, `data.organic_results[7].inline_links.cited_by.total`. Venue: E-1396.

Marked file: 11-Bin-Regularization-MARKED.pdf

Original article · Scholar cited-by list

ARTICLE 12 · SUBSTANTIVE CITATION REPORT

Bayesian Bits: Unifying Quantization and Pruning

Mart van Baalen; Christos Louizos; Markus Nagel; Rana Ali Amjad; Ying Wang; Tijmen Blankevoort; Max Welling
NeurIPS 2020 · 207 Google Scholar citations

ICORE2026 A* venue | >100 Scholar citations

Documented contribution: Credited independent introduction; experimental comparison

Qualcomm AI Research authors credit reference [15], the TQT paper, together with LSQ [8], with having "independently introduced" the learning of the quantization scale along with the model parameters for a fixed-bitwidth network; both papers, they explain, expose the scale parameter to the learning process so that it is optimized jointly with the network's parameters (p. 6). Figure 2b compares Bayesian Bits on MobileNetV2 with TQT, AdaRound and LSQ (p. 8).

Authorship link: Jain, Gural, Wu and Dick (2019), Trained uniform quantization for accurate and efficient neural network inference on fixed-point hardware. Bibliography p. 10.

Reviewed version: Published NeurIPS PDF. **Pinpoint:** PDF/printed pp. 6, 8, scale learning and TQT in Figure 2b and its caption; bibliography p. 10, reference [15].
Count: E-0430, `data.organic_results[9].inline_links.cited_by.total`. Venue: E-0331.

Marked file: 12-Bayesian-Bits-MARKED.pdf

Original article · Scholar cited-by list

SELECTED CITING ARTICLES

ARTICLE 13 · SUBSTANTIVE CITATION REPORT

F8Net: Fixed-Point 8-bit Only Multiplication for Network Quantization

Qing Jin; Jian Ren; Richard Zhuang; Sumant Hanumante; Zhengang Li; Zhiyu Chen; Yanzhi Wang; Kaiyuan Yang; Sergey Tulyakov
ICLR 2022 · 72 Google Scholar citations

ICORE2026 A* venue

Documented contribution: Design positioning and benchmark comparison

Snap Inc., Northeastern University and Rice University researchers define F8Net against the fixed-point quantization of Jain et al. (2019), which uses trainable power-of-two scale factors to replace multiplication with bit shifting, stating that their work "differs from previous efforts (Jain et al., 2019) in three major aspects" (p. 2). They contrast F8Net's on-the-fly fractional length with TQT's directly trained fractional length (p. 7) and compare with TQT in Table 1 for MobileNet V1 and V2b (p. 8).

Authorship link: Jain, Gural, Wu and Dick (2019), Trained quantization thresholds for accurate and efficient fixed-point inference of deep neural networks. Bibliography p. 12.

Reviewed version: arXiv:2202.05239 PDF carrying the ICLR 2022 conference header. **Pinpoint:** Physical PDF pp. 2–3, 7–8: positioning against the power-of-two fixed-point scaling of Jain et al. (2019), contrast with directly trained fractional lengths and TQT rows in Table 1; bibliography p. 12.
Count: E-0381, `data.organic_results[1].inline_links.cited_by.total`. Venue: E-0328; E-1496.

Marked file: 13-F8Net-MARKED.pdf

Original article · Reviewed arXiv version · Scholar cited-by list

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ARTICLE 14 · SUBSTANTIVE CITATION REPORT

Accelerating configuration of Reconfigurable Intelligent Surfaces through a hardware-enhanced deep learning approach

Rubén Padial-Allué; Alberto Martín-Martín; Antonio García; Encarnación Castillo; José David Fernández-Rodríguez; Marcos Baena-Molina; Luis Parrilla
Computers and Electrical Engineering 134 (2026), 111101 · No cited-by count in the saved Google Scholar record

Uses or extends TQT

Documented contribution: Direct methodological use

University of Granada-led researchers use TQT in the quantization-aware training of their AMD DPU implementation for configuring reconfigurable intelligent surfaces, stating "For the quantization strategy, a power-of-two scale quantizer with training quantization threshold (TQT) was used [50]" (Section 3.2, p. 9).

Authorship link: Jain, Gural, Wu and Dick (2020), Trained quantization thresholds for accurate and efficient fixed-point inference of deep neural networks. Bibliography p. 17.

Reviewed version: Open-access Elsevier PDF (CC BY), University of Granada repository copy. **Pinpoint:** Physical PDF p. 9, Section 3.2: power-of-two scale quantizer with TQT for the AMD DPU implementation; bibliography p. 17, reference [50].
Count: E-0385, `data.organic_results[1].inline_links.cited_by.total`.

Marked file: 14-RIS-MARKED.pdf

Original article · Scholar cited-by list

SELECTED CITING ARTICLES

ARTICLE 15 · SUBSTANTIVE CITATION REPORT

FELIX: A Ferroelectric FET Based Low Power Mixed-Signal In-Memory Architecture for DNN Acceleration

Taha Soliman; Nellie Laleni; Tobias Kirchner; Franz Müller; Ashish Shrivastava; Thomas Kämpfe; Andre Guntoro; Norbert Wehn
ACM Transactions on Embedded Computing Systems 21(6), Article 84 (2022) · 50 Google Scholar citations

Uses or extends TQT

Documented contribution: Direct methodological use

Robert Bosch GmbH, Fraunhofer IPMS and University of Kaiserslautern researchers evaluate the classification accuracy of their ferroelectric-FET in-memory accelerator with 8-bit/4-bit (input/weight) quantization "based on the optimizations presented in [17]", the TQT paper (Section 6.2, p. 17).

Authorship link: Jain, Gural, Wu and Dick (2019), Trained quantization thresholds for accurate and efficient fixed-point inference of deep neural networks. Bibliography p. 23.

Reviewed version: ACM Transactions on Embedded Computing Systems publisher PDF. **Pinpoint:** Physical PDF p. 17, Section 6.2: 8-bit/4-bit quantization based on the optimizations in [17]; bibliography p. 23, reference [17].
Count: E-0381, `data.organic_results[4].inline_links.cited_by.total`.

Marked file: 15-FELIX-MARKED.pdf

Original article · Scholar cited-by list

ARTICLE 16 · SUBSTANTIVE CITATION REPORT

HMQ: Hardware Friendly Mixed Precision Quantization Block for CNNs

Hai Victor Habi; Roy H. Jennings; Arnon Netzer
ECCV 2020 · 95 Google Scholar citations

ICORE2026 A* venue

Documented contribution: Quantizer-design connection

Sony Semiconductor Israel researchers cite references [24, 41] for hardware-friendly quantizers that are per-tensor, uniform, symmetric and use power-of-two thresholds, and describe these properties as "an imperative requirement for many efficient edge device hardware implementations" with reference to [24], the TQT paper (p. 2). They state that the quantizer used in HMQ "is similar to the one in [24]" before defining its signed and unsigned forms (p. 4).

Authorship link: Jain, Gural, Wu and Dick (2019), Trained quantization thresholds for accurate and efficient fixed-point inference of deep neural networks. Bibliography p. 17.

Reviewed version: Author arXiv manuscript; ECCV publication verified separately. **Pinpoint:** Physical PDF pp. 1–4, hardware-friendly quantization citations and quantizer definition; bibliography p. 17.
Count: E-0368, `data.organic_results[0].inline_links.cited_by.total`. Venue: E-0585; E-0586.

Marked file: 16-HMQ-MARKED.pdf

Original article · Reviewed arXiv version · Scholar cited-by list

SELECTED CITING ARTICLES

ARTICLE 17 · SUBSTANTIVE CITATION REPORT

Up or Down? Adaptive Rounding for Post-Training Quantization

Markus Nagel; Rana Ali Amjad; Mart van Baalen; Christos Louizos; Tijmen Blankevoort
ICML 2020 · 1,224 Google Scholar citations

ICORE2026 A* venue | >100 Scholar citations

Documented contribution: Related-work attribution

In the related-work discussion of AdaRound, Qualcomm researchers credit Jain et al. and Esser et al. with learning the quantization minimum and maximum ranges during training "so that they do not have to be set manually" (p. 5).

Authorship link: Jain, Gural, Wu and Dick (2019), Trained uniform quantization for accurate and efficient neural network inference on fixed-point hardware. Bibliography p. 10.

Reviewed version: Published PMLR PDF; physical pages retained. **Pinpoint:** Physical PDF p. 5, section 4 Background and related work; bibliography physical p. 10.
Count: E-0430, `data.organic_results[2].inline_links.cited_by.total`. Venue: E-0329.

Marked file: 17-AdaRound-MARKED.pdf

Original article · Scholar cited-by list

ARTICLE 18 · SUBSTANTIVE CITATION REPORT

A White Paper on Neural Network Quantization

Markus Nagel; Marios Fournarakis; Rana Ali Amjad; Yelysei Bondarenko; Mart van Baalen; Tijmen Blankevoort
arXiv / Qualcomm white paper, 2021 · 1,362 Google Scholar citations

>100 Scholar citations

Documented contribution: Technical exposition

Qualcomm's white paper on neural network quantization credits Jain et al., Esser et al. and Bhalgat et al. with using the straight-through estimator to calculate the gradient with respect to the quantization parameters, and then develops the gradient derivation for the scale factor (p. 20). It also cites Jain et al. among works that start quantization-aware training from a pretrained FP32 model, which it calls "common practice in literature" (p. 22).

Authorship link: Jain, Gural, Wu and Dick (2019), Trained uniform quantization for accurate and efficient neural network inference on fixed-point hardware. Bibliography p. 26.

Reviewed version: Reviewed arXiv white paper. **Pinpoint:** Physical PDF pp. 20, 22, gradient derivation and QAT initialization; bibliography p. 26.
Count: E-0430, `data.organic_results[1].inline_links.cited_by.total`.

Marked file: 18-White-Paper-MARKED.pdf

Original article · Scholar cited-by list

SELECTED CITING ARTICLES

ARTICLE 19 · SUBSTANTIVE CITATION REPORT

Bringing AI to edge: From deep learning’s perspective

Di Liu; Hao Kong; Xiangzhong Luo; Weichen Liu; Ravi Subramaniam
Neurocomputing 485 (2022), 297–320 · 247 Google Scholar citations

>100 Scholar citations

Documented contribution: Survey citation

This Neurocomputing survey by Nanyang Technological University researchers, conducted in collaboration with HP Inc. (p. 1), lists TQT among 8-bit quantization techniques for weights and activations in its comparison table (p. 8) and summarizes Jain et al.'s trained uniform quantization method for accurate and efficient inference on fixed-point hardware (p. 9).

Authorship link: Jain, Gural, Wu and Dick (2019), Trained uniform quantization for accurate and efficient neural network inference on fixed-point hardware. Bibliography p. 20.

Reviewed version: arXiv author manuscript; journal publication verified separately. **Pinpoint:** PDF/printed pp. 8–9, TQT comparison-table row and named quantization discussion; bibliography p. 20, reference [121]; affiliations p. 1.
Count: E-0366, `data.organic_results[0].inline_links.cited_by.total`. Venue: E-0352.

Marked file: 19-Bringing-AI-to-Edge-MARKED.pdf

Original article · Reviewed arXiv version · Scholar cited-by list

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ARTICLE 20 · SUBSTANTIVE CITATION REPORT

INT8 Activation Ternary or Binary Weights Networks: Unifying Between INT8 and Lower-bit Width Quantization

Ninnart Fuengfusin; Hakaru Tamukoh
Journal of Robotics, Networking and Artificial Life 9(2) (2022), 171–176 · No cited-by count in the saved Google Scholar record

Uses or extends TQT

Documented contribution: Use of a TQT implementation (Brevitas)

Kyushu Institute of Technology researchers quantize the activations of their ITBWN networks to INT8 with TQT while using binary or ternary weights (pp. 3–4). They describe TQT's removal of the zero point, its power-of-two scale and its threshold training with the straight-through estimator (p. 2), and report that INT8 activations from TQT or BinaryConnect allow the weight-quantized models to deploy without the cost of floating-point accumulation (p. 5). They use the default TQT implementation in Xilinx Brevitas, which they describe as having "minor differences with the TQT settings": no power-of-two scale and a different scale initialization (p. 3).

Authorship link: Jain, Gural, Wu and Dick (2020), Trained quantization thresholds for accurate and efficient fixed-point inference of deep neural networks. Bibliography p. 5.

Reviewed version: Published journal PDF (J-STAGE). **Pinpoint:** Physical PDF pp. 2–5: TQT as the INT8 method, activations quantized with TQT via Xilinx Brevitas, TQT-based settings and conclusion; bibliography p. 5, reference 11.
Count: E-0422, `data.organic_results[9].inline_links.cited_by.total`.

Marked file: 20-ITBWN-MARKED.pdf

Original article · Scholar cited-by list

SELECTED CITING ARTICLES

ARTICLE 21 · SUBSTANTIVE CITATION REPORT

Efficient SAR Vessel Detection for FPGA-Based On-Satellite Sensing

Colin Laganier; Liam Fletcher; Elim Kwan; Richard Walters; Victoria Nockles
SEC '25, Tenth ACM/IEEE Symposium on Edge Computing (2025) · 6 Google Scholar citations

Uses or extends TQT

Documented contribution: Methodological use via Vitis AI

The Alan Turing Institute researchers deploy a YOLOv8-based SAR vessel detector on an AMD/Xilinx Kria KV260 FPGA, quantizing it to INT8 with the Vitis AI Quantizer using symmetric power-of-two scaling; the sentence describing post-training quantization and quantization-aware training (QAT) "using a straight-through estimator" cites reference [35], the TQT paper (p. 5). They retrain with a tuned quantizer-parameter learning rate (Section 4.2.2) and report that the deployed QAT models match the expected GPU performance with little to no degradation (Table 4, Section 4.3, p. 11).

Authorship link: Jain, Gural, Wu and Dick (2020), Trained quantization thresholds for accurate and efficient fixed-point inference of deep neural networks. Bibliography p. 16.

Reviewed version: Published ACM version (open access, CC BY-NC 4.0). **Pinpoint:** Physical PDF p. 5, Section 3.2: INT8 Vitis AI quantization with power-of-two scaling and QAT using a straight-through estimator [35]; p. 11, Section 4.2.2, Table 4 and Section 4.3: QAT with a quantizer parameter learning rate and the deployed FPGA QAT models; bibliography p. 16, reference [35].
Count: E-0386, `data.organic_results[11].inline_links.cited_by.total`.

Marked file: 21-SAR-Vessel-MARKED.pdf

Original article · Scholar cited-by list

Citation coverage across the research portfolio

Entry	Cited work	Profile citations	Records retrieved	Grouped works	Mapped works
01	Trained quantization thresholds for accurate and efficient fixed-point inference of deep neural networks	227	216	206	149
02	Trained uniform quantization for accurate and efficient neural network inference on fixed-point hardware	71	71	69	47
03	Degradation of 2DEG transport properties in GaN-capped AlGaN/GaN heterostructures at 600° C in oxidizing and inert environments	22	22	20	11
04	Interdigitated Pt-GaN Schottky interfaces for high-temperature soot-particulate sensing	15	15	15	5
05	Training a fully convolutional neural network to route integrated Circuits	13	13	13	6
06	A novel self-consistent model based optimal filter design for the improved dynamic performance of 3-phase PLLs for phase tracking under grid imperfections (Part 2)	4	4	4	3
07	System and method for implementing neural networks in integrated circuits	1	1	1	1
08	Gallium Nitride Microelectronics for High-Temperature Environments	1	1	1	1
09	An ML Compilation Stack for Autonomous Vehicles using MLIR	0	0	0	0
10	Systematic DL Optimizations for Accurate and Efficient TensorRT Deployments on Cruise's Self-Driving Fleet	0	0	0	0

A citing work can appear in several publication rows. Unique survey totals group overlapping records and publication versions.

The saved Scholar profile records 354 citations. The inventory retains 323 works from 337 archived Scholar records. Publication affiliations and patent applicant addresses establish locations for 219 works; the remaining 104 have their review status recorded in the inventory.

Dated Scholar responses document discovery and article-level counts. Source PDFs establish the selected discussions and affiliation evidence. The coverage table and inventory provide the denominators for each measure.

Publication-sourced affiliations and research connections

The map presents 232 institution/location entries from 219 grouped citing works in 30 countries/territories. Each pin opens the institutions, citing papers and affiliation-page evidence. Several institutions in one city share a display pin; multi-institution papers contribute several locations.

Affiliations come from the reviewed publication; citing patents are placed at the applicant or assignee address printed on the front page. Explicit campus and city details determine placement; ROR supplies institution-city geography when the publication lists an institution without a city. Hollow markers show country-level affiliations. Coordinates are geographic display anchors.

The map documents the geographic reach of citing authors' publication affiliations and citing patents' applicants, including coauthor- and employer-linked works. Article passages establish methodological adaptation, implementation, comparison, design connections or technical exposition. Joint attribution is retained where several methods are credited.

Research files

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Citation-Inventory.csv: every grouped citing work with its map and review status.
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Substantive Citation Report · Interactive map

Counts refer to the saved October 7, 2026 snapshot. Hashes and source provenance are recorded in the research manifests.